Controller HilandDairy_ChilledWater_System_081325
Hiland Dairy Chilled Water System Program
Controller Fault Handler
Power-Up Handler

Tasks

MainTask
- **MainProgram**
  - **MainRoutine**
- **Buffer_IO**
  - **Main**
  - **HMI_TREND_CONTROL**
    Logic to drive trend visibility in the FT ME Runtime
  - **Slot1_IF8**
  - **Slot2_OF4**
  - **Slot3_OF4**
  - **Slot4_IQ16**
  - **Slot5_OB16**
  - **Slot7_IF8**
  - **Slot8_OF8C**
  - **VFD**
    Drive PlantPax
- **Pump_Control**
  - **Main**
  - **CGSP1**
  - **CGSP2**
  - **CGSP3**
  - **CGTP1**
  - **CWP1**
  - **CWP2**
  - **HHWS_LOGIC**
  - **Test**

Unscheduled

Motion Groups

Ungrouped Axes

Add-On Instructions

- **Input_Debounce**
  - **Logic**
- **Product_Counters**
  - **Logic**
- **P_AIChan**
  Analog Input Channel
  - **Logic**
    Normal Scan - PV is scaled and signal quality checked
  - **Prescan**
    Clear "lurking" Commands, reset internal timers
  - **EnableInFalse**
    Off-Scan: clear "lurking" Commands; reset internal timers; clear Fail Alarm, flag PV as Bad quality
- **P_AIn**
  Analog Input
  - **Logic**
    Normal Scan - PV is scaled and alarms are generated
  - **Prescan**
    Clear Ownership. clear "lurking" Commands
  - **EnableInFalse**
    Off-Scan: no Val update or alarm checking. PV Bad Quality ("stale") is asserted.

P_AInAdv
Advanced Analog Input
Logic
Normal Scan - PV is scaled and alarms are generated
Prescan
Clear Ownership. clear "lurking" Commands
EnableInFalse
Off-Scan: no Val update or alarm checking. PV Bad Quality ("stale") is asserted.

P_AInDual
Dual Analog Input
Logic
Normal Scan - Inputs scaled, PV selected, alarms generated
Prescan
Clear Ownership. clear "lurking" Commands
EnableInFalse
Off-Scan: no Val update or alarm checking. PV Bad Qualtiy ("stale") is asserted.

P_Alarm
Alarm
Logic
Perform Alarm processing: delay, latch/reset, acknowledge, enable/disable, inhibit/uninhibit, suppress/unsuppress
Prescan
Clear "lurking" Commands
EnableInFalse
Perform Alarm processing, but Invert Alarm Input ("Inp") for use on LD rung (EnableIn False)

P_AOut
Analog Output
Logic
Normal Scan: control Analog Output
Prescan
Clear Ownership. clear "lurking" Commands
EnableInFalse
Out of Service is asserted, Output OFF (zero), Alarms inhibited.

P_CmdSrc
Command Source
Logic
Selects Hand, OOS, Maint, Ovrd, Ext, Oper, Prog based on Inputs and Commands
Prescan
Sets internal state and enumerated / bit status to "not initialized".
EnableInFalse
Processes received Commands, but shows "Programmed Out of Service"

P_DIn
Discrete Input
Logic
Normal Scan - status is updated and alarms are generated
Prescan
Clear "lurking" Commands
EnableInFalse
Same as main Logic routine, except Inp is inverted (for use in LD logic)

P_DOut
Discrete Output
Logic
Control Discrete Output, monitoring Feedbacks
Prescan
Clear Ownershp, clear "lurking" Commands, de-energize output
EnableInFalse
Out of Service; output is OFF; feedback Status shown if available; Alarms inhibited.

P_Gate
Condition Gate/Delay
Logic
Gate the input and apply on-delay and off-delay timing
Prescan
Set the Gate Input to 1 (default value) on prescan
EnableInFalse
Same as main Logic routine, except Inp is inverted (for use in LD logic) (Inp_Gate is NOT inverted!!!)

P_Intlk
Interlocks
Logic
Check Interlock Conditions
Prescan
Clear "lurking" Commands
EnableInFalse
Set summary "OK" Status bits to FALSE (0)

P_Logic
Boolean Logic
Logic
Examine Inputs and execute Logic Gates and Output timers
Prescan
Reset On-Delay and Off-Delay timers, clear snapshot timestamp and internal data, clear "lurking" commands
EnableInFalse
Set Output to False and reset On-Delay and Off-Delay output timers

P_Motor
Single Speed Motor
Logic
Normal Scan: control / monitor single speed motor
Prescan
Clear Ownership, clear "lurking" Commands
EnableInFalse
Out of Service is asserted; Outputs OFF; Run Status shown but Alarms inhibited

P_Perm
Permissives
Logic
Check Permissive Conditions
EnableInFalse
Set summary "OK" Status bits to FALSE (0)

P_PF755
PowerFlex 755 Drive
Logic
Normal Scan: control / monitor Variable-Speed Drive
Prescan
Clear Ownership, set drive to STOP, clear "lurking" Commands
EnableInFalse
Out of Service is asserted; Outputs OFF; Run Status shown but Alarms disabled

P_PIDE
Proportional + Integral + Derivative
Logic
Normal Scan: control PID Loop
Prescan
Clear Ownership. clear "lurking" Commands, set powerup Loop Mode, SP, and CV per configuration.
EnableInFalse
Disabled is asserted, output is set to configured Interlock CV, alarms are disabled

P_PIDE_only
Container for PIDE only

Logic
Function Block Diagram routine with bare PIDE block. No Autotune tag, but user can add.

P_RunTime
Run Time and Starts

Logic
Accumulate Run Time (if runnings) and Count Starts

Prescan
Set Current Run time internally to 0. Display value is left in place.

EnableInFalse
False Rung or Enable In False: as if motor is Stopped

P_ValveC
Control Valve

Logic
Normal Scan: manipulate Control Valve with position feedback

Prescan
Clear Ownership. clear "lurking" Commands

EnableInFalse
Out of Service is asserted, Output OFF (zero), Alarms inhibited

P_ValveHO
Hand Operated Valve

Logic
Normal Scan: monitor hand-operated valve

Prescan
Clear "lurking" Commands

EnableInFalse
Out of Service is asserted; Trip Output OFF; Valve Status shown but Alarms inhibited

P_ValveMO
Motor Operated Valve

Logic
Normal Scan: control / monitor motor-operated valve

Prescan
Clear Ownership, set to De-Energized state, clear "lurking" Commands

EnableInFalse
Out of Service is asserted; Outputs OFF; Valve Status shown but Alarms inhibited

P_ValveMP
Mix Proof Valve

Logic
Normal Scan: control / monitor mix proof valve

Prescan
Clear Ownership, set target to Closed state, actual to de-energized state, clear "lurking" Commands

EnableInFalse
Disabled is asserted; Outputs OFF; Valve Status shown based on feedbacks if possible, but Alarms disabled

P_ValveSO
Solenoid Operated Valve

Logic
Normal Scan: control / monitor solenoid-operated valve

Prescan
Clear Ownership, set to De-Energized state, clear "lurking" Commands

EnableInFalse
Out of Service is asserted; Outputs OFF; Valve Status shown but Alarms inhibited.

P_VSD
Variable Speed Drive

Logic
Normal Scan: control / monitor Variable-Speed Drive

Prescan
Clear Ownership, set drive to STOP, clear "lurking" Commands

EnableInFalse
Out of Service is asserted; Outputs OFF; Run Status shown but Alarms inhibited

T_Clock
Date/Time Clock

Logic
Perform clock functions: set, get time, determine shift

Prescan
Clear lurking time setting

Data Types

User-Defined

Alarm_System
Main PLC Alarm Management structure

Auto_REV

Controls

Current

Data

DateTime
Date and TimeOfDay (absolute) or Time (duration/interval)

GC_HMI

Line_Counts

Motor

Photoeye

P_DescList
Code / Description List Entry

P_PF755_Inp
PowerFlex 755 Drive Input Assembly

P_PF755_Out
PowerFlex 755 Drive Output Assembly

P_PFEmb_FltAlmRec
Fault / Alarm Record for PowerFlex VFD Embedded Ethernet Adapter

Speed

Status

System

Strings

STRING_12

STRING_16_UDT

STRING_20

STRING_40

STRING_8

STRING_Area

STRING_Cond

STRING_NavTag
Navigation Tag

Add-On-Defined

Input_Debounce

Product_Counters

P_AIChan
Analog Input Channel

P_AIn
Analog Input

P_AInAdv
Advanced Analog Input

P_AInDual
Dual Analog Input

P_Alarm

Alarm
P_AOut
Analog Output
P_CmdSrc
Command Source
P_DIn
Discrete Input
P_DOut
Discrete Output
P_Gate
Condition Gate/Delay
P_Intlk
Interlocks
P_Logic
Boolean Logic
P_Motor
Single Speed Motor
P_Perm
Permissives
P_PF755
PowerFlex 755 Drive
P_PIDE
Proportional + Integral + Derivative
P_PIDE_only
Container for PIDE only
P_RunTime
Run Time and Starts
P_ValveC
Control Valve
P_ValveHO
Hand Operated Valve
P_ValveMO
Motor Operated Valve
P_ValveMP
Mix Proof Valve
P_ValveSO
Solenoid Operated Valve
P_VSD
Variable Speed Drive
T_Clock
Date/Time Clock

Module-Defined

AB:1769_DI16:I:0
AB:1769_DO16:C:0
AB:1769_DO16:I:0
AB:1769_DO16:O:0
AB:1769_IF8:C:0
AB:1769_IF8:I:0
AB:1769_IF8:O:0
AB:1769_OF4:C:0
AB:1769_OF4:I:0
AB:1769_OF4:O:0
AB:1769_OF8C:C:0
AB:1769_OF8C:I:0
AB:1769_OF8C:O:0
AB:20F_DF4E1FF5:I:0

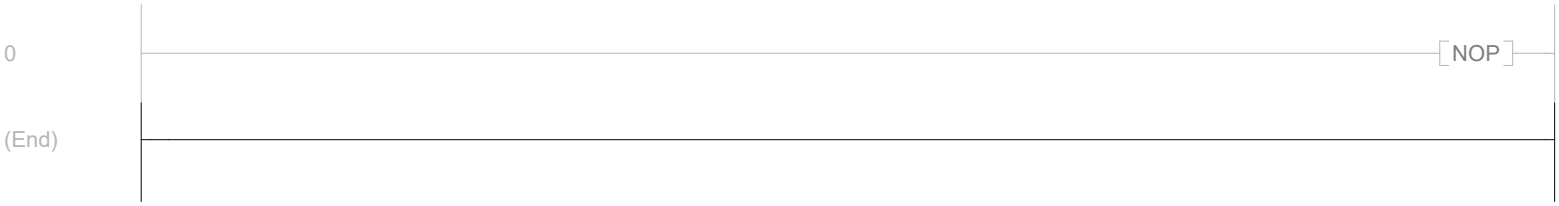
AB:ETHERNET_MODULE:C:0
AB:ETHERNET_MODULE_DINT_256Bytes:I:0
AB:ETHERNET_MODULE_DINT_256Bytes:O:0
AB:PowerFlex753_ENETR_Drive:I:0
AB:PowerFlex753_ENETR_Drive:O:0

Trends

I/O Configuration

1769 Bus
- **[0] 1769-L33ER HilandDairy_ChilledWater_System_081325**
- **[1] 1769-IF8/A Slot_01_IF8**
- **[2] 1769-OF4/A Slot_02_OF4**
- **[3] 1769-OF4/A Slot_03_OF4**
- **[4] 1769-IQ16/A Slot_04_IQ16**
- **[5] 1769-OB16/B Slot_05_OB16**
- **[6] 1769-IF8/A Slot_6_IF8**
- **[7] 1769-OF8C/A Slot_7_OF8C**

Ethernet
- **1769-L33ER HilandDairy_ChilledWater_System_081325**
- **ETHERNET-MODULE RED_LION_DSPLE**
- **PowerFlex 753-ENETR CGSP2_VFD**
  Secondary Pump 2 VFD
- **PowerFlex 753-ENETR CGSP1_VFD**
  Secondary Pump 1 VFD
- **PowerFlex 753-ENETR CGPP1_VFD**
  Primary Punp 1 VFD


0
NOP
(End)

0

Registers for HMI Trend Visibility
TREND_VIS[0]

Number associated with which trend is displayed on the FT ME Runtime

MOV	
Source	1
Dest	TREND 2

1

Registers for HMI Trend Visibility
TREND_VIS[1]

Number associated with which trend is displayed on the FT ME Runtime

MOV	
Source	2
Dest	TREND 2

(End)

0
JSR
Routine Name Slot1_IF8

1
JSR
Routine Name Slot2_OF4

2
JSR
Routine Name Slot3_OF4

3
JSR
Routine Name Slot4_IQ16

4
JSR
Routine Name Slot5_OB16

5
JSR
Routine Name Slot7_IF8

6
JSR
Routine Name Slot8_OF8C

7
Drive PlantPax
JSR
Routine Name VFD

8
Logic to drive trend visibility in the FT ME Runtime
JSR
Routine Name HMI_TREND_CONTROL

(End)

0

Local:1:I.Fault.0

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
CGTP1_TT2.Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:1:I.Ch0Data
-32767
Dest CGTP1_TT2.Inp_PV
-32767.0

Analog Input

P_AIn

P_AIn	CGTP1_TT2 ...
Inp_PV	-32767.0
Cfg_HiHiLim	1.50000000e+038
Cfg_HiLim	1.50000000e+038
Cfg_LoLim	-1.50000000e+038
Cfg_LoLoLim	-1.50000000e+038
Val	-32767.0
Val_InpPV	-32767.0

Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
CGTP1_TT2.PCmd_Reset

1

CV_01
4-20mA
Ball Valve 0-100%
0
CV_01 Position Feedback from Valve (in Raw Input units)

MOV
Source Local:1:I.Ch1Data
-29411
Dest CV1_BELMIO_AO.Inp_PosFdbk
-29447.0

2

CV_01
4-20mA
Ball Valve 0-100%
0
CV_01 Position Feedback from Valve (in Raw Input units)

MOV
Source Local:1:I.Ch2Data
-29401
Dest CV2_BELMIO_AO.Inp_PosFdbk
-29520.0

3

Local:1:I.Fault.3

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
TT1_GLYCOL.Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:1:I.Ch3Data
-32767
Dest TT1_GLYCOL.Inp_PV
-32767.0

Analog Input

P_AIn
P_AIn TT1_GLYCOL ...
Inp_PV -32767.0
Cfg_HiHiLim 1.50000000e+038
Cfg_HiLim 1.50000000e+038
Cfg_LoLim -1.50000000e+038
Cfg_LoLoLim -1.50000000e+038
Val -32767.0
Val_InpPV -32767.0

Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
TT1_GLYCOL.PCmd_Reset

4

Local:1:I.Fault.4

Variable Speed Drive Input Communication Status 0=OK, 1=fail
HHWS1_P1.Inp_IOFault

Variable Speed Drive Speed Feedbak in Drive Units (typ. 0-32767 = 0 to max freq.)

MOV	
Source	Local:1:I.Ch4Data 10119
Dest	HHWS1_P1.Inp_SpeedFdbk 10120.0

Variable Speed Drive

P_VSD	
P_VSD	HHWS1_P1 ...
Ref_FaultCodeList	HHWS1_P1_Fault
Inp_SpeedFdbk	10120.0
Inp_LastFaultCode	0
Inp_Ready	1
Inp_Running	0
Inp_CommandDir	1
Inp_ActualDir	1
Inp_Accelerating	0
Inp_Decelerating	0
Inp_Alarm	0
Inp_Faulted	0
Inp_AtSpeed	0
Inp_FwdPermOK	1
Inp_FwdNBPermOK	1
Inp_RevPermOK	1
Inp_RevNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_SpeedRef	9878.668
Out_Datalink	0.0
Val_SpeedRef	40.0
Val_SpeedFdbk	40.244297
Val_InpDatalink	0.0
Val_OutDatalink	0.0
Val_Fault	0

Out_Stop
Out_Start
Out_Jog
Out_ClearFault
Out_Fwd
Out_Rev
Out_Horn
Sts_Stopped
Sts_StartingFwd
Sts_StartingRev
Sts_RunningFwd
Sts_RunningRev
Sts_StoppingFwd
Sts_StoppingRev
Sts_JoggingFwd
Sts_JoggingRev
Sts_ActualDir
Sts_Alarm
Sts_AtSpeed
Sts_SpeedLimited
Sts_Available
Nrdy_Intlk
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Variable Speed Drive Program Command to Reset drive fault and all Alarms requiring Reset
HHWS1_P1.PCmd_Reset

5

Local:1:I.Fault.5

Variable Speed Drive Input Communication Status 0=OK, 1=fail
HHWS1_P2.Inp_IOFault

Variable Speed Drive Speed Feedbak in Drive Units (typ. 0-32767 = 0 to max freq.)

MOV
Source Local:1:I.Ch5Data
10095
Dest HHWS1_P2.Inp_SpeedFdbk
10096.0

Variable Speed Drive

P_VSD

P_VSD	HHWS1_P2 ...
Ref_FaultCodeList	HHWS1_P2_Fault
Inp_SpeedFdbk	10096.0
Inp_LastFaultCode	0
Inp_Ready	1
Inp_Running	0
Inp_CommandDir	1
Inp_ActualDir	1
Inp_Accelerating	0
Inp_Decelerating	0
Inp_Alarm	0
Inp_Faulted	0
Inp_AtSpeed	0
Inp_FwdPermOK	1
Inp_FwdNBPermOK	1
Inp_RevPermOK	1
Inp_RevNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_SpeedRef	9878.668
Out_Datalink	0.0
Val_SpeedRef	40.0
Val_SpeedFdbk	40.22
Val_InpDatalink	0.0
Val_OutDatalink	0.0
Val_Fault	0

Out_Stop
Out_Start
Out_Jog
Out_ClearFault
Out_Fwd
Out_Rev
Out_Horn
Sts_Stopped
Sts_StartingFwd
Sts_StartingRev
Sts_RunningFwd
Sts_RunningRev
Sts_StoppingFwd
Sts_StoppingRev
Sts_JoggingFwd
Sts_JoggingRev
Sts_ActualDir
Sts_Alarm
Sts_AtSpeed
Sts_SpeedLimited
Sts_Available
Nrdy_Intlk
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Variable Speed Drive Program Command to Reset drive fault and all Alarms requiring Reset
HHWS1_P2.PCmd_Reset

6

Variable Speed Drive Speed Feedbak in Drive Units (typ. 0-32767 = 0 to max freq.)

MOV
Source Local:1:I.Ch6Data
-29784
Dest CTCHW_FAN1.Inp_SpeedFdbk
-29724.0

7

Local:1:I.Fault.7

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
HHWS1_TT1.Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:1:I.Ch7Data
-32767
Dest HHWS1_TT1.Inp_PV
-32767.0

Analog Input

P_AIn
P_AIn HHWS1_TT1 ...
Inp_PV -32767.0
Cfg_HiHiLim 1.50000000e+038
Cfg_HiLim 1.50000000e+038
Cfg_LoLim -1.50000000e+038
Cfg_LoLoLim -1.50000000e+038
Val -32767.0
Val_InpPV -32767.0

Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
HHWS1_TT1.PCmd_Reset

(End)

0

CV_01
4-20mA
Ball Valve 0-100%
0
CV_01

P_ValveC

P_ValveC	CV1_BELMIO_AO	...
Inp_PosFdbk	-29447.0	Out_Open
Inp_OpenLS	0	Out_Close
Inp_ClosedLS	0	Sts_Err
Inp_ActuatorFault	0	Sts_Hand
Inp_IntlkOK	1	Sts_OoS
Inp_NBIntlkOK	1	Sts_Maint
Out_CV	-32767.0	Sts_Ovrd
Val_Pos	5.0660725	Sts_Ext
Val_CVSet	0.0	Sts_Prog
Val_CVOut	0.0	Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

CV_01 4-20mA Ball Valve 0-100% 0 CV_01 Program Command to reset shed latches and cleared alarms
CV1_BELMIO_AO.PCmd_Reset

MOV

Source	CV1_BELMIO_AO.Out_CV -32767.0
Dest	Local:2:O.Ch0Data -32767

1

CV_01
4-20mA
Ball Valve 0-100%
0
CV_01

P_ValveC

P_ValveC	CV2_BELMIO_AO	...
Inp_PosFdbk	-29520.0	Out_Open
Inp_OpenLS	0	Out_Close
Inp_ClosedLS	0	Sts_Err
Inp_ActuatorFault	0	Sts_Hand
Inp_IntlkOK	1	Sts_OoS
Inp_NBIntlkOK	1	Sts_Maint
Out_CV	-32767.0	Sts_Ovrd
Val_Pos	4.95468	Sts_Ext
Val_CVSet	0.0	Sts_Prog
Val_CVOut	0.0	Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

CV_01 4-20mA Ball Valve 0-100% 0 CV_01 Program Command to reset shed latches and cleared alarms
CV2_BELMIO_AO.PCmd_Reset

MOV

Source	CV2_BELMIO_AO.Out_CV -32767.0
Dest	Local:2:O.Ch1Data -32767

2

MOV
Source HHWS1_P1.Out_SpeedRef
9878.668
Dest Local:2:O.Ch2Data
9879

3

MOV
Source HHWS1_P2.Out_SpeedRef
9878.668
Dest Local:2:O.Ch3Data
9879

(End)

0

Variable Speed Drive

P_VSD

P_VSD	CTCHW_FAN1 ...
Ref_FaultCodeList	CTCHW_FAN1_fault
Inp_SpeedFdbk	-29724.0
Inp_LastFaultCode	0
Inp_Ready	1
Inp_Running	0
Inp_CommandDir	1
Inp_ActualDir	1
Inp_Accelerating	0
Inp_Decelerating	0
Inp_Alarm	0
Inp_Faulted	0
Inp_AtSpeed	0
Inp_FwdPermOK	1
Inp_FwdNBPermOK	1
Inp_RevPermOK	1
Inp_RevNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_SpeedRef	0.0
Out_Datalink	0.0
Val_SpeedRef	0.0
Val_SpeedFdbk	0.0
Val_InpDatalink	0.0
Val_OutDatalink	0.0
Val_Fault	0

Out_Stop
Out_Start
Out_Jog
Out_ClearFault
Out_Fwd
Out_Rev
Out_Horn
Sts_Stopped
Sts_StartingFwd
Sts_StartingRev
Sts_RunningFwd
Sts_RunningRev
Sts_StoppingFwd
Sts_StoppingRev
Sts_JoggingFwd
Sts_JoggingRev
Sts_ActualDir
Sts_Alarm
Sts_AtSpeed
Sts_SpeedLimited
Sts_Available
Nrdy_Intlk
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Variable Speed Drive Program Command to Reset drive fault and all Alarms requiring Reset
CTCHW_FAN1.PCmd_Reset

MOV

Source	CTCHW_FAN1.Out_SpeedRef 0.0
Dest	Local:3:O.Ch0Data 0

1

Control Valve

P_ValveC

P_ValveC	CGBM_CV1 ...
Inp_PosFdbk	-32767.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	-32767.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[3].PCmd_Reset

MOV

Source	CGBM_CV1.Out_CV 0.0
Dest	Local:3:O.Ch1Data 0

2

Control Valve

P_ValveC

P_ValveC	SPARE_AO[4] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[4].PCmd_Reset

MOV

Source	SPARE_AO[4].Out_CV 0.0
Dest	Local:3:O.Ch2Data 0

3

Control Valve

P_ValveC	
P_ValveC	SPARE_AO[5] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open
Out_Close
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[5].PCmd_Reset

MOV	
Source	SPARE_AO[5].Out_CV 0.0
Dest	Local:3:O.Ch3Data 0

(End)

0 Local:4:I.Data.0 — C910-485 Heat Trace Alarm Relay HT_ALM_RLY

1 Local:4:I.Data.1 — CT-1 Vibration Cutout Switch CT1_VIB_CO_SW

2 Local:4:I.Data.2 — CT-2 Vibration Cutout Switch CT2_VIB_CO_SW

3 Local:4:I.Data.3 — Feedback Water Pump CWP1_FB

4 Local:4:I.Data.4 — Feedback Water Pump CWP2_FB

5 Local:4:I.Data.5 — Feedback Secondary Pump CGSP1_FB

6 Local:4:I.Data.6 — Feedback Secondary Pump CGSP2_FB

7 Local:4:I.Data.7 — Feedback Secondary Pump CGSP3_FB

8 Local:4:I.Data.8 — Feedback Tertiary Pump CGTP1_FB

9 Local:4:I.Data.9 — Motor Operated Valve Valve Open Limit Switch 1=confirmed OPEN CG_MCV1.Inp_OpenLS

10 Local:4:I.Data.10 — Motor Operated Valve Valve Closed Limit Switch 1=confirmed CLOSED CG_MCV1.Inp_ClosedLS

11 Local:4:I.Data.11 — Motor Operated Valve Valve Open Limit Switch 1=confirmed OPEN CG_MCV2.Inp_OpenLS

12 Local:4:I.Data.12 — Motor Operated Valve Valve Closed Limit Switch 1=confirmed CLOSED CG_MCV2.Inp_ClosedLS

13
Local:4:I.Data.13
Motor Operated Valve Valve Open Limit Switch 1=confirmed OPEN
CG_MCV3.Inp_OpenLS

14
Local:4:I.Data.14
Motor Operated Valve Valve Closed Limit Switch 1=confirmed CLOSED
CG_MCV3.Inp_ClosedLS

15
Local:4:I.Data.15
SPARE[0]

(End)

0

Discrete Output

P_DOut

P_DOut	CTCHW_FAN_HEATER ...
Inp_OnFdbk	0
Inp_OffFdbk	0
Inp_PermOK	1
Inp_NBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1

Out, Out_Horn, Sts_Out, Sts_Pulsing, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Discrete Output
Primary Output 1=ON, 0=OFF
CTCHW_FAN_HEATER.Out

Local:5:O.Data.0

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Discrete Output Program Command to ResetShed Latches and Cleared Alarms
CTCHW_FAN_HEATER.PCmd_Reset

1

Single Speed Motor

P_Motor

P_Motor	CGTP_P1 ...
Inp_RunFdbk	0
Inp_PermOK	1
Inp_NBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1

Out_Run, Out_Horn, Sts_Stopped, Sts_Starting, Sts_Running, Sts_Stopping, Sts_Jogging, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Single Speed Motor
1=Run Motor, 0=Stop Motor
CGTP_P1.Out_Run

Local:5:O.Data.5

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Single Speed Motor Program Command to ResetShed Latches and Cleared Alarms
CGTP_P1.PCmd_Reset

2

Motor Operated Valve

P_ValveMO

P_ValveMO	CG_MCV1 ...
Inp_OpenLS	0
Inp_ClosedLS	1
Inp_ActuatorFault	0
Inp_OpenPermOK	1
Inp_OpenNBPermOK	1
Inp_ClosePermOK	1
Inp_CloseNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1

Out_Open
Out_Close
Out_Horn
Sts_Closed
Sts_CmdToOpen
Sts_Opening
Sts_Opened
Sts_CmdToClose
Sts_Closing
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Motor Operated Valve
1=Run actuator motor
to OPEN Valve
CG_MCV1.Out_Open

Local:5:O.Data.2

Main PLC Alarm
Management structure
Latched Bit for 1
Scan to Reset All
Alarms
Alarm_Global.Reset_Latch

Motor Operated Valve
Program Command to
clear shed latches
and cleared alarms
CG_MCV1.PCmd_Reset

3

Motor Operated Valve

P_ValveMO

P_ValveMO	CG_MCV2 ...
Inp_OpenLS	0
Inp_ClosedLS	1
Inp_ActuatorFault	0
Inp_OpenPermOK	1
Inp_OpenNBPermOK	1
Inp_ClosePermOK	1
Inp_CloseNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1

Out_Open
Out_Close
Out_Horn
Sts_Closed
Sts_CmdToOpen
Sts_Opening
Sts_Opened
Sts_CmdToClose
Sts_Closing
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Motor Operated Valve
1=Run actuator motor
to OPEN Valve
CG_MCV2.Out_Open

Local:5:O.Data.3

Main PLC Alarm
Management structure
Latched Bit for 1
Scan to Reset All
Alarms
Alarm_Global.Reset_Latch

Motor Operated Valve
Program Command to
clear shed latches
and cleared alarms
CG_MCV2.PCmd_Reset

4

Motor Operated Valve

P_ValveMO	
P_ValveMO	CG_MCV3 ...
Inp_OpenLS	0
Inp_ClosedLS	1
Inp_ActuatorFault	0
Inp_OpenPermOK	1
Inp_OpenNBPermOK	1
Inp_ClosePermOK	1
Inp_CloseNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1

Out_Open
Out_Close
Out_Horn
Sts_Closed
Sts_CmdToOpen
Sts_Opening
Sts_Opened
Sts_CmdToClose
Sts_Closing
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Motor Operated Valve 1=Run actuator motor to OPEN Valve
CG_MCV3.Out_Open — Local:5:O.Data.4

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch — Motor Operated Valve Program Command to clear shed latches and cleared alarms
CG_MCV3.PCmd_Reset

5

CGTP-1 Tertiary Pump Motor Start
CGTP1_MS — Local:5:O.Data.1

6

Motorized Control Valve CT1 Open/Close
MCV1_CT1 — Local:5:O.Data.6

7

Motorized Control Valve CT2 Open/Close
MCV2_CT2 — Local:5:O.Data.7

8

Motorized Control Valve Bypass Open/Close
MCV3_BP — Local:5:O.Data.8

9

Variable Speed Drive 1=Start/Run Drive, 0=Stop Drive (for held starter type)
HHWS1_P1.Out_Run — Local:5:O.Data.10

10

Variable Speed Drive 1=Start/Run Drive, 0=Stop Drive (for held starter type)
HHWS1_P2.Out_Run — Local:5:O.Data.11

11
Variable Speed Drive Program Command to Start Drive Forward
CTCHW_FAN1.PCmd_StartFwd
NOP
Local:5:O.Data.12

12
SPARE[4]
Local:5:O.Data.12

13
SPARE[5]
Local:5:O.Data.13

14
Single Speed Motor

P_Motor	
P_Motor	CTCHW_P1 ...
Inp_RunFdbk	0
Inp_PermOK	1
Inp_NBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1

Out_Run
Out_Horn
Sts_Stopped
Sts_Starting
Sts_Running
Sts_Stopping
Sts_Jogging
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Single Speed Motor 1=Run Motor, 0=Stop Motor
CTCHW_P1.Out_Run
Local:5:O.Data.14

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Single Speed Motor Program Command to ResetShed Latches and Cleared Alarms
CTCHW_P1.PCmd_Reset

15
CWP-2 Condensor Water Pump Motor Start
CWP2_MS
Local:5:O.Data.15

(End)

0

Local:6:I.Fault.0

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
HHWS1_PT1.Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:6:I.Ch0Data
-32767
Dest HHWS1_PT1.Inp_PV
-32767.0

Analog Input

P_AIn
P_AIn HHWS1_PT1 ...
Inp_PV -32767.0
Cfg_HiHiLim 1.50000000e+038
Cfg_HiLim 1.50000000e+038
Cfg_LoLim -1.50000000e+038
Cfg_LoLoLim -1.50000000e+038
Val -32767.0
Val_InpPV -32767.0
Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
HHWS1_PT1.PCmd_Reset

1

Local:6:I.Fault.1

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
HHWS1_PT2.Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:6:I.Ch1Data
-32767
Dest HHWS1_PT2.Inp_PV
-32767.0

Analog Input

P_AIn
P_AIn HHWS1_PT2 ...
Inp_PV -32767.0
Cfg_HiHiLim 1.50000000e+038
Cfg_HiLim 1.50000000e+038
Cfg_LoLim -1.50000000e+038
Cfg_LoLoLim -1.50000000e+038
Val -32767.0
Val_InpPV -32767.0
Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
HHWS1_PT2.PCmd_Reset

2

Control Valve Position Feedback from Valve (in Raw Input units)

MOV	
Source	Local:6:I.Ch2Data -32767
Dest	CGBM_CV1.Inp_PosFdbk -32767.0

3

Local:6:I.Fault.3

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
CG_DPT1.Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV	
Source	Local:6:I.Ch3Data -32767
Dest	CG_DPT1.Inp_PV -32767.0

Analog Input

P_AIn	
P_AIn	CG_DPT1 ...
Inp_PV	-32767.0
Cfg_HiHiLim	1.50000000e+038
Cfg_HiLim	1.50000000e+038
Cfg_LoLim	-1.50000000e+038
Cfg_LoLoLim	-1.50000000e+038
Val	-32767.0
Val_InpPV	-32767.0

Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
CG_DPT1.PCmd_Reset

4

Local:6:I.Fault.4

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
SPARE_AI[7].Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:6:I.Ch4Data
-32767
Dest SPARE_AI[7].Inp_PV
-32767.0

Analog Input

P_AIn
P_AIn SPARE_AI[7] ...
Inp_PV -32767.0
Cfg_HiHiLim 1.50000000e+038
Cfg_HiLim 1.50000000e+038
Cfg_LoLim -1.50000000e+038
Cfg_LoLoLim -1.50000000e+038
Val -32767.0
Val_InpPV -32767.0
Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
SPARE_AI[7].PCmd_Reset

5

Local:6:I.Fault.5

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
SPARE_AI[8].Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:6:I.Ch5Data
-32767
Dest SPARE_AI[8].Inp_PV
-32767.0

Analog Input

P_AIn
P_AIn SPARE_AI[8] ...
Inp_PV -32767.0
Cfg_HiHiLim 1.50000000e+038
Cfg_HiLim 1.50000000e+038
Cfg_LoLim -1.50000000e+038
Cfg_LoLoLim -1.50000000e+038
Val -32767.0
Val_InpPV -32767.0
Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
SPARE_AI[8].PCmd_Reset

6

Local:6:I.Fault.6

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
SPARE_AI[9].Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:6:I.Ch6Data
-32767
Dest SPARE_AI[9].Inp_PV
-32767.0

Analog Input

P_AIn

P_AIn	SPARE_AI[9] ...
Inp_PV	-32767.0
Cfg_HiHiLim	1.50000000e+038
Cfg_HiLim	1.50000000e+038
Cfg_LoLim	-1.50000000e+038
Cfg_LoLoLim	-1.50000000e+038
Val	-32767.0
Val_InpPV	-32767.0

Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
SPARE_AI[9].PCmd_Reset

7

Local:6:I.Fault.7

Analog Input 1=PV or I/O Comms Status Bad, 0=OK
SPARE_AI[10].Inp_PVBad

Analog Input Input Signal (process variable) from Sensor

MOV
Source Local:6:I.Ch7Data
-32767
Dest SPARE_AI[10].Inp_PV
-32767.0

Analog Input

P_AIn

P_AIn	SPARE_AI[10] ...
Inp_PV	-32767.0
Cfg_HiHiLim	1.50000000e+038
Cfg_HiLim	1.50000000e+038
Cfg_LoLim	-1.50000000e+038
Cfg_LoLoLim	-1.50000000e+038
Val	-32767.0
Val_InpPV	-32767.0

Sts_Err
Sts_HiHi
Sts_Hi
Sts_Lo
Sts_LoLo
Sts_Fail
Sts_FailShelved

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarm_Global.Reset_Latch

Analog Input Program Command to Reset all Alarms requiring Reset
SPARE_AI[10].PCmd_Reset

(End)

0

Control Valve

P_ValveC	
P_ValveC	SPARE_AO[6] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[6].PCmd_Reset

MOV	
Source	SPARE_AO[6].Out_CV 0.0
Dest	Local:7:O.Ch0Data 0

1

Control Valve

P_ValveC	
P_ValveC	SPARE_AO[7] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[7].PCmd_Reset

MOV	
Source	SPARE_AO[7].Out_CV 0.0
Dest	Local:7:O.Ch1Data 0

2

Control Valve

P_ValveC	
P_ValveC	SPARE_AO[8] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[8].PCmd_Reset

MOV	
Source	SPARE_AO[8].Out_CV 0.0
Dest	Local:7:O.Ch2Data 0

3

Control Valve

P_ValveC	
P_ValveC	SPARE_AO[9] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[9].PCmd_Reset

MOV	
Source	SPARE_AO[9].Out_CV 0.0
Dest	Local:7:O.Ch3Data 0

4

Control Valve

P_ValveC		
P_ValveC	SPARE_AO[10] ...	
Inp_PosFdbk	0.0	Out_Open
Inp_OpenLS	0	Out_Close
Inp_ClosedLS	0	Sts_Err
Inp_ActuatorFault	0	Sts_Hand
Inp_IntlkOK	1	Sts_OoS
Inp_NBIntlkOK	1	Sts_Maint
Out_CV	0.0	Sts_Ovrd
Val_Pos	0.0	Sts_Ext
Val_CVSet	0.0	Sts_Prog
Val_CVOut	0.0	Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[10].PCmd_Reset

MOV	
Source	SPARE_AO[10].Out_CV 0.0
Dest	Local:7:O.Ch4Data 0

5

Control Valve

P_ValveC		
P_ValveC	SPARE_AO[11] ...	
Inp_PosFdbk	0.0	Out_Open
Inp_OpenLS	0	Out_Close
Inp_ClosedLS	0	Sts_Err
Inp_ActuatorFault	0	Sts_Hand
Inp_IntlkOK	1	Sts_OoS
Inp_NBIntlkOK	1	Sts_Maint
Out_CV	0.0	Sts_Ovrd
Val_Pos	0.0	Sts_Ext
Val_CVSet	0.0	Sts_Prog
Val_CVOut	0.0	Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[11].PCmd_Reset

MOV	
Source	SPARE_AO[11].Out_CV 0.0
Dest	Local:7:O.Ch5Data 0

6

Control Valve

P_ValveC	
P_ValveC	SPARE_AO[12] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[12].PCmd_Reset

MOV	
Source	SPARE_AO[12].Out_CV 0.0
Dest	Local:7:O.Ch6Data 0

7

Control Valve

P_ValveC	
P_ValveC	SPARE_AO[13] ...
Inp_PosFdbk	0.0
Inp_OpenLS	0
Inp_ClosedLS	0
Inp_ActuatorFault	0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Out_CV	0.0
Val_Pos	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Out_Open, Out_Close, Sts_Err, Sts_Hand, Sts_OoS, Sts_Maint, Sts_Ovrd, Sts_Ext, Sts_Prog, Sts_Oper

Main PLC Alarm Management structure Latched Bit for 1 Scan to Reset All Alarms
Alarms_Global.Reset_Latch

Control Valve Program Command to reset shed latches and cleared alarms
SPARE_AO[13].PCmd_Reset

MOV	
Source	SPARE_AO[13].Out_CV 0.0
Dest	Local:7:O.Ch7Data 0

(End)

0

PowerFlex 755 Drive

P_PF755

P_PF755	CGPP1_VFD ...
Inp	CGPP1_VFD_inp
Out	CGPP1_VFD_out
Ref_GetFaultMSG	CGPP1_VFD_msg ...
Ref_GetFaultDest	CGPP1_VFD_fault
Inp_FwdPermOK	1
Inp_FwdNBPermOK	1
Inp_RevPermOK	1
Inp_RevNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Val_SpeedRef	40.0
Val_SpeedFdbk	40.0

Out_Horn
Sts_Stopped
Sts_StartingFwd
Sts_StartingRev
Sts_RunningFwd
Sts_RunningRev
Sts_StoppingFwd
Sts_StoppingRev
Sts_JoggingFwd
Sts_JoggingRev
Sts_ActualDir
Sts_Alarm
Sts_AtSpeed
Sts_SpeedLimited
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

PowerFlex 755 Drive
Speed Feedback
(actual) from drive

MOV
Source CGPP1_VFD:I.Feedback
40.0
Dest CGPP1_VFD.Val_SpeedFdbk
40.0

MOV
Source CGPP1_VFD.Val_SpeedRef
40.0
Dest CGPP1_VFD:O.Reference
40.0

PowerFlex 755 Drive
Input Assembly

MOV
Source CGPP1_VFD:I.DriveStatus
2#0010_0000_0010_0001_0000_0101_0000_1111
Dest CGPP1_VFD_inp.DriveStatus
2#0010_0000_0010_0001_0000_0101_0000_1111

MOV
Source CGPP1_VFD_out.LogicCommand
2#0000_0000_0000_0000_0000_0000_0001_0000
Dest CGPP1_VFD:O.LogicCommand
2#0000_0000_0000_0000_0000_0000_0001_0000

1

PowerFlex 755 Drive

P_PF755

P_PF755	CGSP1_VFD ...
Inp	CGSP1_VFD_inp
Out	CGSP1_VFD_out
Ref_GetFaultMSG	CGSP1_VFD_msg ...
Ref_GetFaultDest	CGSP1_VFD_fault
Inp_FwdPermOK	1
Inp_FwdNBPermOK	1
Inp_RevPermOK	1
Inp_RevNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Val_SpeedRef	40.0
Val_SpeedFdbk	40.0

Out_Horn
Sts_Stopped
Sts_StartingFwd
Sts_StartingRev
Sts_RunningFwd
Sts_RunningRev
Sts_StoppingFwd
Sts_StoppingRev
Sts_JoggingFwd
Sts_JoggingRev
Sts_ActualDir
Sts_Alarm
Sts_AtSpeed
Sts_SpeedLimited
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

PowerFlex 755 Drive
Speed Feedback
(actual) from drive

MOV
Source CGSP1_VFD:I.Feedback
40.0
Dest CGSP1_VFD.Val_SpeedFdbk
40.0

MOV
Source CGSP1_VFD.Val_SpeedRef
40.0
Dest CGSP1_VFD:O.Reference
40.0

PowerFlex 755 Drive
Input Assembly

MOV
Source CGSP1_VFD:I.DriveStatus
2#0010_0000_0010_0001_0000_0101_0000_1111
Dest CGSP1_VFD_inp.DriveStatus
2#0010_0000_0010_0001_0000_0101_0000_1111

MOV
Source CGSP1_VFD_out.LogicCommand
2#0000_0000_0000_0000_0000_0000_0001_0000
Dest CGSP1_VFD:O.LogicCommand
2#0000_0000_0000_0000_0000_0000_0001_0000

2

PowerFlex 755 Drive

P_PF755	
P_PF755	CGSP2_VFD ...
Inp	CGSP2_VFD_inp
Out	CGSP2_VFD_out
Ref_GetFaultMSG	CGSP2_VFD_msg ...
Ref_GetFaultDest	CGSP2_VFD_fault
Inp_FwdPermOK	1
Inp_FwdNBPermOK	1
Inp_RevPermOK	1
Inp_RevNBPermOK	1
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Val_SpeedRef	40.0
Val_SpeedFdbk	40.0

Out_Horn
Sts_Stopped
Sts_StartingFwd
Sts_StartingRev
Sts_RunningFwd
Sts_RunningRev
Sts_StoppingFwd
Sts_StoppingRev
Sts_JoggingFwd
Sts_JoggingRev
Sts_ActualDir
Sts_Alarm
Sts_AtSpeed
Sts_SpeedLimited
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

PowerFlex 755 Drive
Speed Feedback
(actual) from drive

MOV	
Source	CGSP2_VFD:I.Feedback
	40.0
Dest	CGSP2_VFD.Val_SpeedFdbk
	40.0

MOV	
Source	CGSP2_VFD.Val_SpeedRef
	40.0
Dest	CGSP2_VFD:O.Reference
	40.0

PowerFlex 755 Drive
Input Assembly

MOV	
Source	CGSP2_VFD:I.DriveStatus
	2#0010_0000_0010_0001_0000_0101_0000_1111
Dest	CGSP2_VFD_inp.DriveStatus
	2#0010_0000_0010_0001_0000_0101_0000_1111

MOV	
Source	CGSP2_VFD_out.LogicCommand
	2#0000_0000_0000_0000_0000_0000_0001_0000
Dest	CGSP2_VFD:O.LogicCommand
	2#0000_0000_0000_0000_0000_0000_0001_0000

(End)

0

NOP

AUTO / MANUAL SELECTION

1

Auto Push Button
CGSP1.Control.Auto_PB

CGSP1.Status.Auto

CGSP1.Status.Auto

Manual Push Button
CGSP1.Control.Manual_PB

2

CGSP1.Status.Auto

CGSP1.Status.Manual

Fault Handling

3

CGSP1.Control.HMI_Reset_PB

CGSP1.Control.Reset

System.Control.Reset

4

Start Request
CGSP1.Control.Start_Req

CGSP1.Status.Permissives_OK

CGSP1.Status.Running

Manual Start Request
CGSP1.Control.Manual_Start

CGSP1.Status.Faulted

CGSP1.Control.Reset

TON
Timer CGSP1.Fault_TMR
Preset 10000
Accum 0
EN
DN

CGSP1.Fault_TMR.DN

CGSP1.Status.Faulted

Permissives

5

CGSP1.Status.Faulted

CGSP1.Status.Permissives_OK

MTR Start / Stop

6

CGSP1.Status.Manual

CGSP1.Control.HMI_Start_PB

Stop Command
CGSP1.Control.Stop_Cmd

Manual Start Request
CGSP1.Control.Manual_Start

Manual Start Request
CGSP1.Control.Manual_Start

7

CGSP1.Status.Auto
CGSP1.Control.HMI_Start_PB
Stop Command
CGSP1.Control.Stop_Cmd
Start Request
CGSP1.Control.Start_Req

Start Request
CGSP1.Control.Start_Req

8

Start Request
CGSP1.Control.Start_Req
CGSP1.Status.Permissives_OK
CGSP-1 Secondary Pump Motor Start
CGSP1_MS

Manual Start Request
CGSP1.Control.Manual_Start

9

CGSP1.Status.Auto
Stop Command
System.Control.Stop_Cmd
Stop Command
CGSP1.Control.Stop_Cmd

CGSP1.Status.Manual
System.Control.HMI_Stop_PB

CGSP1.Control.HMI_Stop_PB

CGSP1.Status.Faulted

Stop Command
CGSP1.Control.Stop_Cmd
CGSP1.Control.Reset

Running Status

10

Feedback Secondary Pump
CGSP1_FB
CGSP1.Status.Running

11

Manual Start Request
CGSP1.Control.Manual_Start
Stop Command
CGSP1.Control.Stop_Cmd
PowerFlex 755 Drive Program Command to Start Drive Forward
CGSP1_VFD.PCmd_StartFwd

Start Request
CGSP1.Control.Start_Req

(End)

0

NOP

AUTO / MANUAL SELECTION

1

Auto Push Button
CGSP2.Control.Auto_PB

CGSP2.Status.Auto

CGSP2.Status.Auto

Manual Push Button
CGSP2.Control.Manual_PB

2

CGSP2.Status.Auto

CGSP2.Status.Manual

Fault RESET

3

CGSP2.Control.HMI_Reset_PB

CGSP2.Control.Reset

System.Control.Reset

Fault Handling

4

Start Request
CGSP2.Control.Start_Req

CGSP2.Status.Permissives_OK

CGSP2.Status.Running

Manual Start Request
CGSP2.Control.Manual_Start

CGSP2.Status.Faulted

CGSP2.Control.Reset

TON
Timer CGSP2.Fault_TMR
Preset 10000
Accum 0
EN
DN

CGSP2.Fault_TMR.DN

CGSP2.Status.Faulted

Permissives

5

CGSP2.Status.Faulted

CGSP2.Status.Permissives_OK

MTR Start / Stop

6

CGSP2.Status.Manual

CGSP2.Control.HMI_Start_PB

Stop Command
CGSP2.Control.Stop_Cmd

Manual Start Request
CGSP2.Control.Manual_Start

Manual Start Request
CGSP2.Control.Manual_Start

7

CGSP2.Status.Auto
CGSP2.Control.HMI_Start_PB
Stop Command
CGSP2.Control.Stop_Cmd
Start Request
CGSP2.Control.Start_Req

Start Request
CGSP2.Control.Start_Req

8

Start Request
CGSP2.Control.Start_Req
CGSP2.Status.Permissives_OK
CGSP-2 Secondary Pump Motor Start
CGSP2_MS

Manual Start Request
CGSP2.Control.Manual_Start

9

CGSP2.Status.Auto
Stop Command
System.Control.Stop_Cmd
Stop Command
CGSP2.Control.Stop_Cmd

CGSP2.Status.Manual
System.Control.HMI_Stop_PB

CGSP2.Control.HMI_Stop_PB

CGSP2.Status.Faulted

Stop Command
CGSP2.Control.Stop_Cmd
CGSP2.Control.Reset

10

Running Status
Feedback Secondary Pump
CGSP2_FB
CGSP2.Status.Running

(End)

0

NOP

AUTO / MANUAL SELECTION

1

Auto Push Button
CGSP3.Control.Auto_PB

CGSP3.Status.Auto

CGSP3.Status.Auto

Manual Push Button
CGSP3.Control.Manual_PB

2

CGSP3.Status.Auto

CGSP3.Status.Manual

Fault RESET

3

CGSP3.Control.HMI_Reset_PB

CGSP3.Control.Reset

System.Control.Reset

Fault Handling

4

Start Request
CGSP3.Control.Start_Req

CGSP3.Status.Permissives_OK

CGSP3.Status.Running

Manual Start Request
CGSP3.Control.Manual_Start

CGSP3.Status.Faulted

CGSP3.Control.Reset

TON
Timer CGSP3.Fault_TMR
Preset 10000
Accum 0
EN
DN

CGSP3.Fault_TMR.DN

CGSP3.Status.Faulted

Permissives

5

CGSP3.Status.Faulted

CGSP3.Status.Permissives_OK

MTR Start / Stop

6

CGSP3.Status.Manual

CGSP3.Control.HMI_Start_PB

Stop Command
CGSP3.Control.Stop_Cmd

Manual Start Request
CGSP3.Control.Manual_Start

Manual Start Request
CGSP3.Control.Manual_Start

7

CGSP3.Status.Auto
CGSP3.Control.HMI_Start_PB
Stop Command
CGSP3.Control.Stop_Cmd
Start Request
CGSP3.Control.Start_Req

Start Request
CGSP3.Control.Start_Req

8

Start Request
CGSP3.Control.Start_Req
CGSP3.Status.Permissives_OK
CGSP-3 Secondary Pump Motor Start
CGSP3_MS

Manual Start Request
CGSP3.Control.Manual_Start

9

CGSP3.Status.Auto
Stop Command
System.Control.Stop_Cmd
Stop Command
CGSP3.Control.Stop_Cmd

CGSP3.Status.Manual
System.Control.HMI_Stop_PB

CGSP3.Control.HMI_Stop_PB

CGSP3.Status.Faulted

Stop Command
CGSP3.Control.Stop_Cmd
CGSP3.Control.Reset

10

Running Status
Feedback Secondary Pump
CGSP3_FB
CGSP3.Status.Running

(End)

0

NOP

AUTO / MANUAL SELECTION

1

Auto Push Button
CGTP1.Control.Auto_PB

CGTP1.Status.Auto

CGTP1.Status.Auto

Manual Push Button
CGTP1.Control.Manual_PB

2

CGTP1.Status.Auto

CGTP1.Status.Manual

Fault RESET

3

CGTP1.Control.HMI_Reset_PB

CGTP1.Control.Reset

System.Control.Reset

Fault Handling

4

Start Request
CGTP1.Control.Start_Req

CGTP1.Status.Permissives_OK

CGTP1.Status.Running

Manual Start Request
CGTP1.Control.Manual_Start

CGTP1.Status.Faulted

CGTP1.Control.Reset

TON
Timer CGTP1.Fault_TMR
Preset 10000
Accum 0
EN
DN

CGTP1.Fault_TMR.DN

CGTP1.Status.Faulted

Permissives

5

CGTP1.Status.Faulted

CGTP1.Status.Permissives_OK

MTR Start / Stop

6

CGTP1.Status.Manual

CGTP1.Control.HMI_Start_PB

Stop Command
CGTP1.Control.Stop_Cmd

Manual Start Request
CGTP1.Control.Manual_Start

Manual Start Request
CGTP1.Control.Manual_Start

7

CGTP1.Status.Auto
CGTP1.Control.HMI_Start_PB
Stop Command
CGTP1.Control.Stop_Cmd
Start Request
CGTP1.Control.Start_Req

Start Request
CGTP1.Control.Start_Req

8

Start Request
CGTP1.Control.Start_Req
CGTP1.Status.Permissives_OK
CGTP-1 Tertiary Pump Motor Start
CGTP1_MS

Manual Start Request
CGTP1.Control.Manual_Start

9

CGTP1.Status.Auto
Stop Command
System.Control.Stop_Cmd
Stop Command
CGTP1.Control.Stop_Cmd

CGTP1.Status.Manual
System.Control.HMI_Stop_PB

CGTP1.Control.HMI_Stop_PB

CGTP1.Status.Faulted

Stop Command
CGTP1.Control.Stop_Cmd
CGTP1.Control.Reset

10

Running Status
Feedback Tertiary Pump
CGTP1_FB
CGTP1.Status.Running

(End)

0

NOP

AUTO / MANUAL SELECTION

1

Auto Push Button
CWP1.Control.Auto_PB

CWP1.Status.Auto

CWP1.Status.Auto

Manual Push Button
CWP1.Control.Manual_PB

2

CWP1.Status.Auto

CWP1.Status.Manual

Fault RESET

3

CWP1.Control.HMI_Reset_PB

CWP1.Control.Reset

System.Control.Reset

Fault Handling

4

Start Request
CWP1.Control.Start_Req

CWP1.Status.Permissives_OK

CWP1.Status.Running

Manual Start Request
CWP1.Control.Manual_Start

CWP1.Status.Faulted

CWP1.Control.Reset

TON
Timer CWP1.Fault_TMR
Preset 10000
Accum 0
EN
DN

CWP1.Fault_TMR.DN

CWP1.Status.Faulted

Permissives

5

CWP1.Status.Faulted

CWP1.Status.Permissives_OK

MTR Start / Stop

6

CWP1.Status.Manual

CWP1.Control.HMI_Start_PB

Stop Command
CWP1.Control.Stop_Cmd

Manual Start Request
CWP1.Control.Manual_Start

Manual Start Request
CWP1.Control.Manual_Start

7

CWP1.Status.Auto
CWP1.Control.HMI_Start_PB
Stop Command
CWP1.Control.Stop_Cmd
Start Request
CWP1.Control.Start_Req

Start Request
CWP1.Control.Start_Req

8

Start Request
CWP1.Control.Start_Req
CWP1.Status.Permissives_OK
CWP-1 Condensor Water Pump Motor Start
CWP1_MS

Manual Start Request
CWP1.Control.Manual_Start

9

CWP1.Status.Auto
Stop Command
System.Control.Stop_Cmd
Stop Command
CWP1.Control.Stop_Cmd

CWP1.Status.Manual
System.Control.HMI_Stop_PB

CWP1.Control.HMI_Stop_PB

CWP1.Status.Faulted

Stop Command
CWP1.Control.Stop_Cmd
CWP1.Control.Reset

10

Running Status
Feedback Water Pump
CWP1_FB
CWP1.Status.Running

(End)

0

NOP

AUTO / MANUAL SELECTION

1

Auto Push Button
CWP2.Control.Auto_PB

CWP2.Status.Auto

CWP2.Status.Auto

Manual Push Button
CWP2.Control.Manual_PB

2

CWP2.Status.Auto

CWP2.Status.Manual

Fault RESET

3

CWP2.Control.HMI_Reset_PB

CWP2.Control.Reset

System.Control.Reset

Fault Handling

4

Start Request
CWP2.Control.Start_Req

CWP2.Status.Permissives_OK

CWP2.Status.Running

Manual Start Request
CWP2.Control.Manual_Start

CWP2.Status.Faulted

CWP2.Control.Reset

TON
Timer CWP2.Fault_TMR
Preset 10000
Accum 0
EN
DN

CWP2.Fault_TMR.DN

CWP2.Status.Faulted

Permissives

5

CWP2.Status.Faulted

CWP2.Status.Permissives_OK

MTR Start / Stop

6

CWP2.Status.Manual

CWP2.Control.HMI_Start_PB

Stop Command
CWP2.Control.Stop_Cmd

Manual Start Request
CWP2.Control.Manual_Start

Manual Start Request
CWP2.Control.Manual_Start

7

CWP2.Status.Auto
CWP2.Control.HMI_Start_PB
Stop Command
CWP2.Control.Stop_Cmd
Start Request
CWP2.Control.Start_Req
Start Request
CWP2.Control.Start_Req

8

Start Request
CWP2.Control.Start_Req
CWP2.Status.Permissives_OK
CWP-2 Condensor Water Pump Motor Start
CWP2_MS
Manual Start Request
CWP2.Control.Manual_Start

9

CWP2.Status.Auto
Stop Command
System.Control.Stop_Cmd
Stop Command
CWP2.Control.Stop_Cmd
CWP2.Status.Manual
System.Control.HMI_Stop_PB
CWP2.Control.HMI_Stop_PB
CWP2.Status.Faulted
Stop Command
CWP2.Control.Stop_Cmd
CWP2.Control.Reset

10

Running Status
Feedback Water Pump
CWP2_FB
CWP2.Status.Running

(End)

0

HHWS1_HMI_Start

HHWS1_START_REQ

1

HHWS1_START_REQ

HHWS1_PumpSel_1

Analog Input Input Signal (process variable) from Sensor

GRT
Source A HHWS1_PT1.Inp_PV -32767.0
Source B HHWS1_HMI_PT_SP 0

Variable Speed Drive Program Setting of Speed Reference (EU)

MOV
Source HHWS1_HMI_SpeedRef 0
Dest HHWS1_P1.PSet_SpeedRef 40.0

Variable Speed Drive Program Command to Start Drive Forward

HHWS1_P1.PCmd_StartFwd

HHWS1_PumpSEl_2

Analog Input Input Signal (process variable) from Sensor

GRT
Source A HHWS1_PT1.Inp_PV -32767.0
Source B HHWS1_HMI_PT_SP 0

Variable Speed Drive Program Setting of Speed Reference (EU)

MOV
Source HHWS1_HMI_SpeedRef 0
Dest HHWS1_P2.PSet_SpeedRef 40.0

Variable Speed Drive Program Command to Start Drive Forward

HHWS1_P2.PCmd_StartFwd

2

Proportional + Integral + Derivative

P_PIDE

P_PIDE	HHWS1_PIDE ...
Inp_PV	0.0
Inp_CascSP	0.0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Inp_Hand	0
Out_CV	0.0
Val_PV	0.0
Val_SP	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Sts_Casc
Sts_Auto
Sts_Man
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Proportional + Integral + Derivative Process Variable (PVEU)

MOV

Source	HHWS1_PT1.Inp_PV -32767.0
Dest	HHWS1_PIDE.Inp_PV 0.0

Proportional + Integral + Derivative Value of SP being used (after ramping and clamping (PVEU)

MOV

Source	HHWS1_HMI_PT_SP 0
Dest	HHWS1_PIDE.Val_SP 0.0

MOV

Source	HHWS1_PIDE.Out_CV 0.0
Dest	HHWS1_HMI_SpeedRef 0

3

Proportional + Integral + Derivative

P_PIDE

P_PIDE	HHWS1_TEMP_PIDE ...
Inp_PV	0.0
Inp_CascSP	0.0
Inp_IntlkOK	1
Inp_NBIntlkOK	1
Inp_Hand	0
Out_CV	0.0
Val_PV	0.0
Val_SP	0.0
Val_CVSet	0.0
Val_CVOut	0.0

Sts_Casc
Sts_Auto
Sts_Man
Sts_Err
Sts_Hand
Sts_OoS
Sts_Maint
Sts_Ovrd
Sts_Ext
Sts_Prog
Sts_Oper

Proportional + Integral + Derivative Process Variable (PVEU)

MOV

Source	HHWS1_TT1.Inp_PV -32767.0
Dest	HHWS1_TEMP_PIDE.Inp_PV 0.0

Proportional + Integral + Derivative Value of SP being used (after ramping and clamping (PVEU)

MOV

Source	HHWS1_Temp_SP 0
Dest	HHWS1_TEMP_PIDE.Val_SP 0.0

MOV

Source	HHWS1_TEMP_PIDE.Out_CV 0.0
Dest	HHWS1_Valve_OUTCV 0

4

HHWS1_START_REQ

HHWS1_ValveSel_1

Analog Input Analog Input Value (actual, before Substitute PV selection)

GRT
Source A HHWS1_TT1.Val_InpPV
-32767.0
Source B HHWS1_Temp_SP
0

CV_01
4-20mA
Ball Valve 0-100%
0
CV_01 Program Setting of Controlled Variable (output) (CV EU)

MOV
Source HHWS1_Valve_OUTCV
0
Dest CV1_BELMIO_AO.PSet_CV
0.0

Variable Speed Drive Program Command to Start Drive Forward
HHWS1_P1.PCmd_StartFwd

HHWS1_ValveSel_2

Analog Input Analog Input Value (actual, before Substitute PV selection)

GRT
Source A HHWS1_TT1.Val_InpPV
-32767.0
Source B HHWS1_Temp_SP
0

CV_01
4-20mA
Ball Valve 0-100%
0
CV_01 Program Setting of Controlled Variable (output) (CV EU)

MOV
Source HHWS1_Valve_OUTCV
0
Dest CV2_BELMIO_AO.PSet_CV
0.0

Variable Speed Drive Program Command to Start Drive Forward
HHWS1_P2.PCmd_StartFwd

(End)

0 JSR Routine Name CGSP1

1 JSR Routine Name CGSP2

2 JSR Routine Name CGSP3

3 JSR Routine Name CGTP1

4 JSR Routine Name CWP1

5 JSR Routine Name CWP2

(End)

V2 Example

0
NOP

1
Auto Push Button
CWP1.Control.Auto_PB
CWP1.Status.Auto
Manual Push Button
CWP1.Control.Manual_PB
CWP1.Status.Auto
CWP1.Status.Auto

2
CWP1.Status.Auto
CWP1.Status.Manual

3
CWP1.Control.HMI_Reset_PB
System.Control.Reset
CWP1.Control.Reset

4
Start Request
CWP1.Control.Start_Req
CWP1.Status.Permissives_OK
CWP1.Status.Running
Manual Start Request
CWP1.Control.Manual_Start
CWP1.Status.Faulted
CWP1.Control.Reset
TON
Timer CWP1.Fault_TMR
Preset 10000
Accum 0
EN
DN
CWP1.Fault_TMR.DN
CWP1.Status.Faulted

5
CWP1.Status.Faulted
CWP1.Status.Permissives_OK

6
CWP1.Status.Manual
CWP1.Control.HMI_Start_PB
Manual Start Request
CWP1.Control.Manual_Start
Stop Command
CWP1.Control.Stop_Cmd
Manual Start Request
CWP1.Control.Manual_Start

7
CWP1.Status.Auto
CWP1.Control.HMI_Start_PB
Start Request
CWP1.Control.Start_Req
Stop Command
CWP1.Control.Stop_Cmd
Start Request
CWP1.Control.Start_Req

8
Start Request
CWP1.Control.Start_Req
CWP1.Status.Permissives_OK
Manual Start Request
CWP1.Control.Manual_Start
Start Command
CWP1.Control.Start_Cmd

9
CWP1.Status.Auto
Stop Command
System.Control.Stop_Cmd
CWP1.Status.Manual
System.Control.HMI_Stop_PB
CWP1.Control.HMI_Stop_PB
CWP1.Status.Faulted
Stop Command
CWP1.Control.Stop_Cmd
CWP1.Control.Reset
Stop Command
CWP1.Control.Stop_Cmd

10
Feedback Water Pump
CWP1_FB
CWP1.Status.Running

V1 Example

11
Start Command
CGSP1.Control.Start_Cmd
CGSP1.Status.Faulted
CGSP-1 Secondary Pump Motor Start
CGSP1_MS

12
CGSP-1 Secondary Pump Motor Start
CGSP1_MS
Feedback Secondary Pump
CGSP1_FB
CGSP1.Control.Reset
CGSP1.Status.Faulted
TON
Timer CGSP1.Fault_TMR
Preset 10000
Accum 0
EN
DN
CGSP1.Fault_TMR.DN
CGSP1.Status.Faulted

13
Feedback Secondary Pump
CGSP1_FB
CGSP1.Status.Running

(End)

1769 Bus : Local Modules

Local: [0] 1769-L33ER HilandDairy_ChilledWater_System_081325

Hiland Dairy Chilled Water System Program

Type:	1769-L33ER CompactLogix™ 5370 Controller	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	0	Electronic Keying:	Exact Match
Revision:	34.11	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off

Local: [1] 1769-IF8/A Slot_01_IF8

Type:	1769-IF8/A 8 Channel Current/Voltage Analog Input	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	1	Electronic Keying:	Compatible Module
Revision:	2.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	No		

Module Defined Configuration Tag	Value	Data Type
Local:1:C		AB:1769_IF8:C:0
.RTSInterval	100	INT
.RTSEn	0	BOOL
.Ch0Filter	0	SINT
.Ch0AlarmInterruptEn	0	BOOL
.Ch0AlarmLatchEn	0	BOOL
.Ch0AlarmEn	0	BOOL
.Ch0En	1	BOOL
.Ch0Range	3	SINT
.Ch0DataFormat	0	SINT
.Ch0HAlarmLimit	0	INT
.Ch0LAlarmLimit	0	INT
.Ch0AlarmDeadband	0	INT
.Ch1Filter	0	SINT
.Ch1AlarmInterruptEn	0	BOOL
.Ch1AlarmLatchEn	0	BOOL
.Ch1AlarmEn	0	BOOL
.Ch1En	1	BOOL
.Ch1Range	2	SINT
.Ch1DataFormat	0	SINT
.Ch1HAlarmLimit	0	INT
.Ch1LAlarmLimit	0	INT
.Ch1AlarmDeadband	0	INT
.Ch2Filter	0	SINT
.Ch2AlarmInterruptEn	0	BOOL
.Ch2AlarmLatchEn	0	BOOL
.Ch2AlarmEn	0	BOOL
.Ch2En	1	BOOL
.Ch2Range	2	SINT
.Ch2DataFormat	0	SINT
.Ch2HAlarmLimit	0	INT
.Ch2LAlarmLimit	0	INT
.Ch2AlarmDeadband	0	INT
.Ch3Filter	0	SINT

	0	BOOL
.Ch3AlarmInterruptEn		
.Ch3AlarmLatchEn	0	BOOL
.Ch3AlarmEn	0	BOOL
.Ch3En	1	BOOL
.Ch3Range	3	SINT
.Ch3DataFormat	0	SINT
.Ch3HAlarmLimit	0	INT
.Ch3LAlarmLimit	0	INT
	0	INT
.Ch3AlarmDeadband		
.Ch4Filter	0	SINT
	0	BOOL
.Ch4AlarmInterruptEn		
.Ch4AlarmLatchEn	0	BOOL
.Ch4AlarmEn	0	BOOL
.Ch4En	1	BOOL
.Ch4Range	2	SINT
.Ch4DataFormat	0	SINT
.Ch4HAlarmLimit	0	INT
.Ch4LAlarmLimit	0	INT
	0	INT
.Ch4AlarmDeadband		
.Ch5Filter	0	SINT
	0	BOOL
.Ch5AlarmInterruptEn		
.Ch5AlarmLatchEn	0	BOOL
.Ch5AlarmEn	0	BOOL
.Ch5En	1	BOOL
.Ch5Range	2	SINT
.Ch5DataFormat	0	SINT
.Ch5HAlarmLimit	0	INT
.Ch5LAlarmLimit	0	INT
	0	INT
.Ch5AlarmDeadband		
.Ch6Filter	0	SINT
	0	BOOL
.Ch6AlarmInterruptEn		
.Ch6AlarmLatchEn	0	BOOL
.Ch6AlarmEn	0	BOOL
.Ch6En	1	BOOL
.Ch6Range	2	SINT
.Ch6DataFormat	0	SINT
.Ch6HAlarmLimit	0	INT
.Ch6LAlarmLimit	0	INT
	0	INT
.Ch6AlarmDeadband		
.Ch7Filter	0	SINT
	0	BOOL
.Ch7AlarmInterruptEn		
.Ch7AlarmLatchEn	0	BOOL
.Ch7AlarmEn	0	BOOL
.Ch7En	1	BOOL
.Ch7Range	3	SINT
.Ch7DataFormat	0	SINT
.Ch7HAlarmLimit	0	INT
.Ch7LAlarmLimit	0	INT
	0	INT
.Ch7AlarmDeadband		

Local: [2] 1769-OF4/A Slot_02_OF4

Type:	1769-OF4/A 4 Channel Current/Voltage Analog Output	Parent:	Local

Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	2	Electronic Keying:	Compatible Module
Revision:	1.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	No		

Module Defined Configuration Tag	Value	Data Type
Local:2:C		AB:1769_OF4:C:0
.Ch0LimitAlarmLatchEn	0	BOOL
.Ch0OutputEn	1	BOOL
.Ch0RangeType	2	SINT
.Ch0DataFormat	0	SINT
.Ch0HClampValue	32767	INT
.Ch0LClampValue	-32767	INT
.Ch1LimitAlarmLatchEn	0	BOOL
.Ch1OutputEn	1	BOOL
.Ch1RangeType	2	SINT
.Ch1DataFormat	0	SINT
.Ch1HClampValue	32767	INT
.Ch1LClampValue	-32767	INT
.Ch2LimitAlarmLatchEn	0	BOOL
.Ch2OutputEn	1	BOOL
.Ch2RangeType	2	SINT
.Ch2DataFormat	0	SINT
.Ch2HClampValue	32767	INT
.Ch2LClampValue	-32767	INT
.Ch3LimitAlarmLatchEn	0	BOOL
.Ch3OutputEn	1	BOOL
.Ch3RangeType	2	SINT
.Ch3DataFormat	0	SINT
.Ch3HClampValue	32767	INT
.Ch3LClampValue	-32767	INT

Local: [3] 1769-OF4/A Slot_03_OF4

Type:	1769-OF4/A 4 Channel Current/Voltage Analog Output	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	3	Electronic Keying:	Compatible Module
Revision:	1.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	No		

Module Defined Configuration Tag	Value	Data Type
Local:3:C		AB:1769_OF4:C:0
.Ch0LimitAlarmLatchEn	0	BOOL
.Ch0OutputEn	1	BOOL
.Ch0RangeType	2	SINT
.Ch0DataFormat	0	SINT
.Ch0HClampValue	32767	INT

.Ch0LClampValue	-32767	INT
.Ch1LimitAlarmLatchEn	0	BOOL
.Ch1OutputEn	1	BOOL
.Ch1RangeType	3	SINT
.Ch1DataFormat	0	SINT
.Ch1HClampValue	32767	INT
.Ch1LClampValue	-32767	INT
.Ch2LimitAlarmLatchEn	0	BOOL
.Ch2OutputEn	1	BOOL
.Ch2RangeType	3	SINT
.Ch2DataFormat	0	SINT
.Ch2HClampValue	32767	INT
.Ch2LClampValue	-32767	INT
.Ch3LimitAlarmLatchEn	0	BOOL
.Ch3OutputEn	1	BOOL
.Ch3RangeType	3	SINT
.Ch3DataFormat	0	SINT
.Ch3HClampValue	32767	INT
.Ch3LClampValue	-32767	INT

Local: [4] 1769-IQ16/A Slot_04_IQ16

Type:	1769-IQ16/A 16 Point 24V DC Input, Sink/Source	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	4	Electronic Keying:	Compatible Module
Revision:	1.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	No		

Local: [5] 1769-OB16/B Slot_05_OB16

Type:	1769-OB16/B 16 Point 24V DC Output, Source	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	5	Electronic Keying:	Compatible Module
Revision:	2.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	No		

Module Defined Configuration Tag	Value	Data Type
Local:5:C		AB:1769_DO16:C:0
.Config	2#0000_0000_0000_0000	INT
.ProgToFaultEn	0	BOOL
.ProgMode	2#0000_0000_0000_0000	INT
.ProgValue	2#0000_0000_0000_0000	INT
.FaultMode	2#0000_0000_0000_0000	INT
.FaultValue	2#0000_0000_0000_0000	INT

Local: [6] 1769-IF8/A Slot_6_IF8

Type:	1769-IF8/A 8 Channel Current/Voltage Analog Input	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	6	Electronic Keying:	Compatible Module

Revision:	2.2	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	No		

Module Defined Configuration Tag	Value	Data Type
Local:6:C		AB:1769_IF8:C:0
.RTSInterval	100	INT
.RTSEn	0	BOOL
.Ch0Filter	0	SINT
	0	BOOL
.Ch0AlarmInterruptEn		
.Ch0AlarmLatchEn	0	BOOL
.Ch0AlarmEn	0	BOOL
.Ch0En	1	BOOL
.Ch0Range	3	SINT
.Ch0DataFormat	0	SINT
.Ch0HAlarmLimit	0	INT
.Ch0LAlarmLimit	0	INT
	0	INT
.Ch0AlarmDeadband		
.Ch1Filter	0	SINT
	0	BOOL
.Ch1AlarmInterruptEn		
.Ch1AlarmLatchEn	0	BOOL
.Ch1AlarmEn	0	BOOL
.Ch1En	1	BOOL
.Ch1Range	3	SINT
.Ch1DataFormat	0	SINT
.Ch1HAlarmLimit	0	INT
.Ch1LAlarmLimit	0	INT
	0	INT
.Ch1AlarmDeadband		
.Ch2Filter	0	SINT
	0	BOOL
.Ch2AlarmInterruptEn		
.Ch2AlarmLatchEn	0	BOOL
.Ch2AlarmEn	0	BOOL
.Ch2En	1	BOOL
.Ch2Range	3	SINT
.Ch2DataFormat	0	SINT
.Ch2HAlarmLimit	0	INT
.Ch2LAlarmLimit	0	INT
	0	INT
.Ch2AlarmDeadband		
.Ch3Filter	0	SINT
	0	BOOL
.Ch3AlarmInterruptEn		
.Ch3AlarmLatchEn	0	BOOL
.Ch3AlarmEn	0	BOOL
.Ch3En	1	BOOL
.Ch3Range	3	SINT
.Ch3DataFormat	0	SINT
.Ch3HAlarmLimit	0	INT
.Ch3LAlarmLimit	0	INT
	0	INT
.Ch3AlarmDeadband		
.Ch4Filter	0	SINT
	0	BOOL
.Ch4AlarmInterruptEn		
.Ch4AlarmLatchEn	0	BOOL
.Ch4AlarmEn	0	BOOL
.Ch4En	1	BOOL

.Ch4Range	3	SINT
.Ch4DataFormat	0	SINT
.Ch4HAlarmLimit	0	INT
.Ch4LAlarmLimit	0	INT
.Ch4AlarmDeadband	0	INT
.Ch5Filter	0	SINT
.Ch5AlarmInterruptEn	0	BOOL
.Ch5AlarmLatchEn	0	BOOL
.Ch5AlarmEn	0	BOOL
.Ch5En	1	BOOL
.Ch5Range	3	SINT
.Ch5DataFormat	0	SINT
.Ch5HAlarmLimit	0	INT
.Ch5LAlarmLimit	0	INT
.Ch5AlarmDeadband	0	INT
.Ch6Filter	0	SINT
.Ch6AlarmInterruptEn	0	BOOL
.Ch6AlarmLatchEn	0	BOOL
.Ch6AlarmEn	0	BOOL
.Ch6En	1	BOOL
.Ch6Range	3	SINT
.Ch6DataFormat	0	SINT
.Ch6HAlarmLimit	0	INT
.Ch6LAlarmLimit	0	INT
.Ch6AlarmDeadband	0	INT
.Ch7Filter	0	SINT
.Ch7AlarmInterruptEn	0	BOOL
.Ch7AlarmLatchEn	0	BOOL
.Ch7AlarmEn	0	BOOL
.Ch7En	1	BOOL
.Ch7Range	3	SINT
.Ch7DataFormat	0	SINT
.Ch7HAlarmLimit	0	INT
.Ch7LAlarmLimit	0	INT
.Ch7AlarmDeadband	0	INT

Local: [7] 1769-OF8C/A Slot_7_OF8C

Type:	1769-OF8C/A 8 Channel Single Ended Current Analog Output	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	7	Electronic Keying:	Compatible Module
Revision:	2.2	Status:	Standby
Module Fault:	Offline	Inhibit Flag	Off
Use Unicast:	No		

Module Defined Configuration Tag	Value	Data Type
Local:7:C		AB:1769_OF8C:C:0
.Ch0ProgToFaultEn	0	BOOL
.Ch0HoldForInit	0	BOOL
.Ch0ProgMode	0	BOOL
.Ch0FaultMode	0	BOOL
.Ch0RampEn	0	BOOL
.Ch0AlarmLatchEn	0	BOOL

	0	BOOL
.Ch0OverRangeInterruptEn		
	0	BOOL
.Ch0UnderRangeInterruptEn		
.Ch0En	1	BOOL
.Ch0Range	1	SINT
.Ch0DataFormat	0	SINT
.Ch0FaultValue	0	INT
.Ch0ProgValue	0	INT
.Ch0LClampValue	0	INT
.Ch0HClampValue	1	INT
.Ch0RampRate	0	INT
.Ch1ProgToFaultEn	0	BOOL
.Ch1HoldForInit	0	BOOL
.Ch1ProgMode	0	BOOL
.Ch1FaultMode	0	BOOL
.Ch1RampEn	0	BOOL
.Ch1AlarmLatchEn	0	BOOL
	0	BOOL
.Ch1OverRangeInterruptEn		
	0	BOOL
.Ch1UnderRangeInterruptEn		
.Ch1En	1	BOOL
.Ch1Range	1	SINT
.Ch1DataFormat	0	SINT
.Ch1FaultValue	0	INT
.Ch1ProgValue	0	INT
.Ch1LClampValue	0	INT
.Ch1HClampValue	1	INT
.Ch1RampRate	0	INT
.Ch2ProgToFaultEn	0	BOOL
.Ch2HoldForInit	0	BOOL
.Ch2ProgMode	0	BOOL
.Ch2FaultMode	0	BOOL
.Ch2RampEn	0	BOOL
.Ch2AlarmLatchEn	0	BOOL
	0	BOOL
.Ch2OverRangeInterruptEn		
	0	BOOL
.Ch2UnderRangeInterruptEn		
.Ch2En	1	BOOL
.Ch2Range	1	SINT
.Ch2DataFormat	0	SINT
.Ch2FaultValue	0	INT
.Ch2ProgValue	0	INT
.Ch2LClampValue	0	INT
.Ch2HClampValue	1	INT
.Ch2RampRate	0	INT
.Ch3ProgToFaultEn	0	BOOL
.Ch3HoldForInit	0	BOOL
.Ch3ProgMode	0	BOOL
.Ch3FaultMode	0	BOOL
.Ch3RampEn	0	BOOL
.Ch3AlarmLatchEn	0	BOOL
	0	BOOL
.Ch3OverRangeInterruptEn		

	0	BOOL
.Ch3UnderRangeInterruptEn		
.Ch3En	1	BOOL
.Ch3Range	1	SINT
.Ch3DataFormat	0	SINT
.Ch3FaultValue	0	INT
.Ch3ProgValue	0	INT
.Ch3LClampValue	0	INT
.Ch3HClampValue	1	INT
.Ch3RampRate	0	INT
.Ch4ProgToFaultEn	0	BOOL
.Ch4HoldForInit	0	BOOL
.Ch4ProgMode	0	BOOL
.Ch4FaultMode	0	BOOL
.Ch4RampEn	0	BOOL
.Ch4AlarmLatchEn	0	BOOL
	0	BOOL
.Ch4OverRangeInterruptEn		
	0	BOOL
.Ch4UnderRangeInterruptEn		
.Ch4En	1	BOOL
.Ch4Range	1	SINT
.Ch4DataFormat	0	SINT
.Ch4FaultValue	0	INT
.Ch4ProgValue	0	INT
.Ch4LClampValue	0	INT
.Ch4HClampValue	1	INT
.Ch4RampRate	0	INT
.Ch5ProgToFaultEn	0	BOOL
.Ch5HoldForInit	0	BOOL
.Ch5ProgMode	0	BOOL
.Ch5FaultMode	0	BOOL
.Ch5RampEn	0	BOOL
.Ch5AlarmLatchEn	0	BOOL
	0	BOOL
.Ch5OverRangeInterruptEn		
	0	BOOL
.Ch5UnderRangeInterruptEn		
.Ch5En	1	BOOL
.Ch5Range	1	SINT
.Ch5DataFormat	0	SINT
.Ch5FaultValue	0	INT
.Ch5ProgValue	0	INT
.Ch5LClampValue	0	INT
.Ch5HClampValue	1	INT
.Ch5RampRate	0	INT
.Ch6ProgToFaultEn	0	BOOL
.Ch6HoldForInit	0	BOOL
.Ch6ProgMode	0	BOOL
.Ch6FaultMode	0	BOOL
.Ch6RampEn	0	BOOL
.Ch6AlarmLatchEn	0	BOOL
	0	BOOL
.Ch6OverRangeInterruptEn		
	0	BOOL
.Ch6UnderRangeInterruptEn		
.Ch6En	1	BOOL

.Ch6Range	1	SINT
.Ch6DataFormat	0	SINT
.Ch6FaultValue	0	INT
.Ch6ProgValue	0	INT
.Ch6LClampValue	0	INT
.Ch6HClampValue	1	INT
.Ch6RampRate	0	INT
.Ch7ProgToFaultEn	0	BOOL
.Ch7HoldForInit	0	BOOL
.Ch7ProgMode	0	BOOL
.Ch7FaultMode	0	BOOL
.Ch7RampEn	0	BOOL
.Ch7AlarmLatchEn	0	BOOL
.Ch7OverRangeInterruptEn	0	BOOL
.Ch7UnderRangeInterruptEn	0	BOOL
.Ch7En	1	BOOL
.Ch7Range	1	SINT
.Ch7DataFormat	0	SINT
.Ch7FaultValue	0	INT
.Ch7ProgValue	0	INT
.Ch7LClampValue	0	INT
.Ch7HClampValue	1	INT
.Ch7RampRate	0	INT

Ethernet : Local Modules

ETHERNET-MODULE RED_LION_DSPLE

Type:	ETHERNET-MODULE Generic Ethernet Module	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
IP Address or Host Name:	192.168.1.20	Electronic Keying:	Disabled
Revision:	1.1	Status:	Standby
Module Fault:	Offline	Inhibit Flag	On
RPI:	10 ms	Use Unicast:	Yes

Module Defined Configuration Tag	Value	Data Type
RED_LION_DSPLE:C		AB:ETHERNET_MODULE:C:0
.Data		SINT[400]

PowerFlex 753-ENETR CGSP2_VFD

Secondary Pump 2 VFD

Type:	PowerFlex 753-ENETR AC Drive via ENETR	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	0	IP Address or Host Name:	192.168.1.30
Electronic Keying:	Compatible Module	Revision:	20.3
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	RPI:	20 ms
Use Unicast:	Yes		

PowerFlex 753-ENETR CGSP1_VFD

Secondary Pump 1 VFD

Type:	PowerFlex 753-ENETR AC Drive via ENETR	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	0	IP Address or Host Name:	192.168.1.31
Electronic Keying:	Compatible Module	Revision:	20.3
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	RPI:	20 ms
Use Unicast:	Yes		

PowerFlex 753-ENETR CGPP1_VFD

Primary Punp 1 VFD

Type:	PowerFlex 753-ENETR AC Drive via ENETR	Parent:	Local
Vendor:	Rockwell Automation/Allen-Bradley	Vendor ID:	1
Slot:	0	IP Address or Host Name:	192.168.1.32
Electronic Keying:	Compatible Module	Revision:	20.3
Status:	Standby	Module Fault:	Offline
Inhibit Flag	Off	RPI:	20 ms
Use Unicast:	Yes		

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